
5. Nano Devices and Quantum Computing

Introduction - quantum confinement - quantum structures: quantum wells, wires and dots - band gap of nanomaterials. Tunneling - Single electron phenomena: Coulomb blockade - resonant-tunneling diode - single electron transistor - quantum cellular automata - Quantum system for information processing - quantum states - classical bits - quantum bits or qubits - CNOT gate - multiple qubits - Bloch sphere - quantum gates - advantage of quantum computing over classical computing.

INTRODUCTION

A **nanometre** (nm) is one billionth $\left(\frac{1}{10^9}\right)$ of a metre. For comparison, thickness of a single human hair is about 80,000 nm (80 μm), a red blood cell is approximately 7,000 nm (7 μm) wide and a water molecule is almost 0.3 nm across.

Scientists and engineers are nowadays interested in nanoscale which is from 1 nm to 100 nm. At nanoscale, the properties of materials are very different from those at larger scale. Therefore, the nano-world is in between quantum world and macro world.

Nanoscience

It is concerned with the study of phenomena and manipulation of materials at nanometre scale.

Nanotechnology

It is the design, characterization, production and application of structures, devices and systems by controlling shape and size at the nanometre scale.

Nanotechnology means making use of the unique physical properties of atoms, molecules and other materials measuring roughly 1 to 100 nanometre.

The word “nano” comes from nanos, a Greek word meaning dwarf.

Presently, we are using many devices made of nanoelectronic devices. The microelectronics industry was born out of the invention of the bi-polar transistor in 1947 and by the invention of the integrated circuit (IC) in 1958.

Gordon Moore (co-founder of INTEL Corporation) observed that the number of transistors per square inch on IC chip roughly doubled by every 18 to 24 months. This general rule of thumb is now called as **“Moore’s law”**.

By 1960, the minimum feature size of a transistor was approximately 100 μm . At present, manufacturing technology is at transistor size of 22 nm.

Because of the diminishing feature size of transistors and other components, we can say that the electronics industry is already “doing” nanotechnology.

Nanomaterials

Definition

Nanophase materials are newly developed materials with grain size at the nanometre range (10^{-9} m), i.e., in the order of 1 - 100 nm. The particle size in a nano material is 1 - 100 nm. They are simply called nanomaterials.

Different forms of Nanomaterials

Nano-structured material

The structures whose characteristic variations in design length is at the nanoscale.

Nano particles

The particles size in the order of 10^{-9} m are called nanoparticles.

Nano dots

Nanoparticles which consist of homogeneous material, especially those that are almost spherical or cubical in shape.

Nanorods

Nanostructures which are shaped like long sticks or rods with diameter in nanoscale and a length very much longer.

Nanotubes

Nanotubes are nanoscale materials that has a tube like (hollow cylinder) structure.

Nanowires

Nanowires are solid rod-like material with diameter of few nanometres or less.

Fullerenes

A form of carbon having a large molecule consisting of an empty cage of 60 or more carbon atoms.

Nanocomposites

Composite structures whose characteristic dimensions are found at nanoscale.

Cluster

A collection of units (atoms or reactive molecules) upto few tens of units.

Colloids

A stable liquid phase containing particles in the 1-1000 nm range.

Nano electronics

Nanoelectronics refers to the use of nanotechnology in electronic components, especially transistors.

It often refers to transistor devices that are so small that inter-atomic interactions and quantum mechanical properties need to be studied extensively.

Besides, being small and allowing more transistors to be packed into a single chip, the uniform and symmetrical structure of nanotubes allows a higher electron mobility, a symmetrical electron/hole characteristic.

Need for Nanotechnology in Electronics

Today microelectronics are used to solve most of the problems.

The two exceptional disadvantages of micro electronics are:

- Physical size
- Increasing cost of fabrication of integrated circuits.

To overcome these disadvantages, nanotechnology is used.

Advantages of Using Nanotechnology in Electronics

- Increasing the density of memory chips.
- Decreasing the weight and thickness of the screens.
- Nanolithography is used for fabrication of chips.
- Reducing the size of transistors used in integrated circuits.
- Improving display screens on electronic devices.
- Reducing power consumption.

5.1 QUANTUM CONFINEMENT

Definition

It is a process of reduction of the size of the solid such that the energy levels inside become discrete.

In this case, small “**droplets**” of isolated electrons are created. Thus, the energy of a small volume of such materials **are** quantized just like in an atom. This type of artificial or fake atoms have **tunable** electrical properties.

The quantum confinement effect is observed when the size of the particle is too small to be comparable to the wavelength (de-Broglie wavelength) of the electron.

It is to be noted that only a small percentage of electrons that are free to move during confinement and majority of electrons still tightly bound within inner orbitals.

Usually in order to reduce the dimensions of a given volume, either bottom-up approach or top-down approach is followed.

In bottom-up approach, low-volume structures are built atom by atom. In the top-down approach, material is removed from one or more of three dimensions (length, width, height) of a larger solid.

In both cases, a structure small enough for quantum behavior to manifest can be produced.

5.2 QUANTUM STRUCTURES : Quantum wells, wires and dots

Definition

When a bulk material is reduced in its size, atleast one of its dimension, in the order of few nanometres, then the structure is known as quantum structure.

Explanation

The volume of a box can be reduced by shortening its length, width or, and height. The same is true for the region occupied by the electrons in a solid.

There are three dimensions to confine the bulk material. The quantum confinement needs confining at least one of these dimensions to less than 100 nanometers or even just a few nanometers.

The more the dimensions are confined, the more the density of states function looks like that of an atom. This progressive **discretization** gives new ways to understand real atoms, behavior of electrons and developing quantum confined electronic devices.

A structure in which the motion of the electrons or holes are confined in one or more directions by potential barriers is called quantum confined structure.

The quantum confined structure is classified into three types based on the confinement directions. They are

- (i) **quantum well**
- (ii) **quantum wire**
- (iii) **quantum dot**

(i) Quantum well (2 - dimension)

Definition

When the electrons are confined inside a region of minimal width, ie., confinement in one dimension “**quantum-well**” is created.

In other words, if one dimension is reduced to the nanometre range while the other two dimensions remain large, then we get a structure known as **quantum well**.

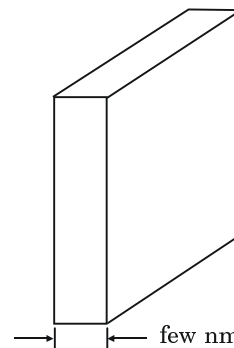


Fig. 5.1 quantum Well (2D)

Fig. 5.1 shows a quantum well or 2-D structure

Construction

Quantum wells are made from alternative layers of different semiconductors or by deposition of very thin metal films.

Explanation

The quantum well is a larger structure in which the carrier particles are free to move in two dimensions. The particles are confined in one dimension hence, they are considered to be **quantum confinement**.

Due to the confinement of carriers, the quantum well structure has important applications in making useful devices.

Use

Quantum wells are now widely used to make semiconductor lasers and other important devices.

(ii) Quantum wire (1 dimension)**Definition**

When the electrons are confined in two mutually perpendicular directions, then the structure is known as quantum wire.

In other words, if two dimensions are reduced and one remains large, the resulting structure is quantum wire.



Fig. 5.2 Quantum wire (1D)

Fig. 5.2 shows a quantum wire or 1-D structure

Explanation

The carriers trapped in such structures are considered to be in 1-D quantum confinement. In this case, a carrier is only free to move its trajectory along the wire.

Example

Examples of quantum wire structures are nanowires, nanorod and nanotube.

(iii) Quantum Dots (0 - dimension)

Definition

When all the three dimensions are minimized the resulting structure is known as quantum dot.

The dot can be particle located inside a larger structure or on its surface. It can also be a place where electrons have been trapped using electrical fields.

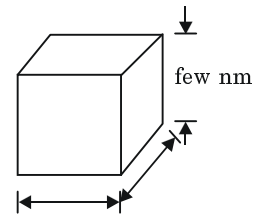


Fig. 5.3 Quantum dot

Fig. 5.3 shows a quantum dot or 0-D structure

Explanation

In quantum dot, the carriers has only confined states i.e., there are no freely moving carriers. Though a quantum dot has many thousands of atoms, carriers it is considered more like a single atom due to its peculiar properties.

Use

Quantum dot can be used in a quantum computer and lasers etc.

Fig. 5.4 and Fig. 5.5 explains the processes of diminishing the size in the case of rectilinear geometry and curvilinear geometry, respectively.

Fig. 5.6 shows a comparison of three quantum confined structures with bulk material.

As quantum wells and quantum wires each have at least one dimension in which the electrons are free to move and these

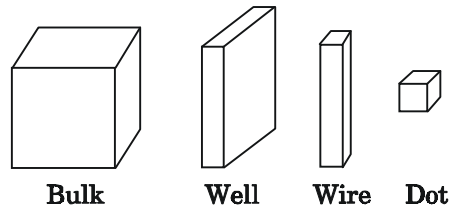


Fig. 5.4 Progressive generation of rectangular nanostructures

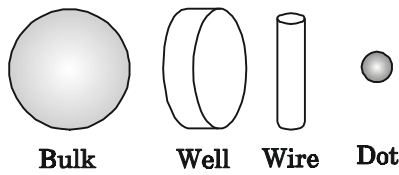


Fig. 5.5 Progressive generation of curvilinear nanostructures

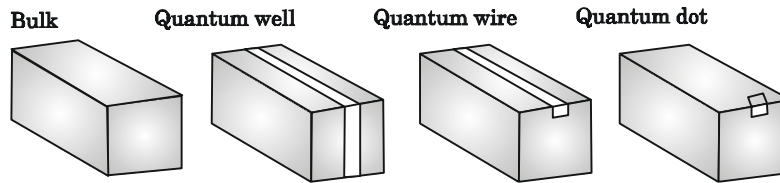


Fig. 5.6 Three quantum structures

structures exhibit “**partial confinement**”. However, quantum dots exhibit “**total confinement**”.

The classification of quantum confined structures is shown in **Table 5.1**.

Table 5.1

Structure	Quantum confinement directions	Number of unconfined dimensions
Bulk	0	3
Quantum well	1	2
Quantum wire	2	1
Quantum dot	3	0

5.3 BANDGAP OF NANOMATERIALS

The electronic properties of metals and semiconductors are determined by their electronic band structure. The band structure changes with particle size. Molecular orbitals get converted into delocalized band states as shown in fig. 5.7.

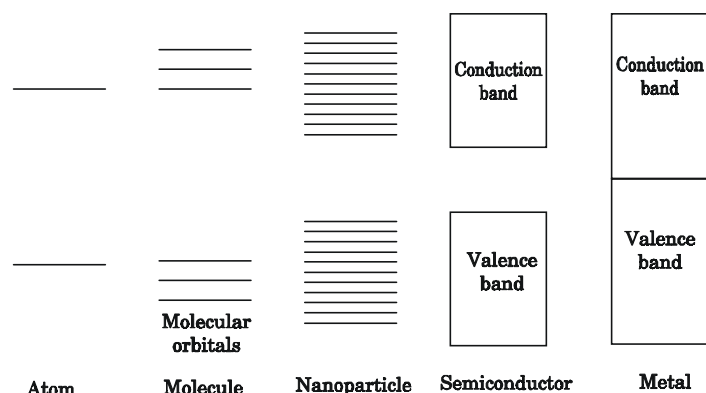


Fig. 5.7 Electronic state transition with a change in size

The band structure of nanocrystals lies between the discrete density of states as in atoms and molecules and continuous band as in crystals.

As the size of the material decreases, the energy separation between the adjacent levels increases. This size quantization effect is responsible for the transition of electronic states from a bulk metal or semiconductor to nanoparticles.

The particles that show this size quantization effect are called Q - particles or quantum dots.

In case of the particle size being less than the de Broglie wavelength, charge carriers can be quantum-mechanically understood as particles in a box and the size of the box can provide the dimensions of the particle.

With a decrease in particle size of metals, the quasi-continuous density of states splits into discrete electronic levels with an increase in the spacing between these levels.

Quantum size effect is most significant for semiconductor nanoparticles. In semiconductor, a bandgap already exists in the bulk state. It also increases and the energy bands gradually convert into discrete molecular electronic levels with a decrease in particle size.

As the size of metal nanoparticles decreases, they tend to lose their metallic character and become semiconductors.

In metals, the quantum size effect exists but it can be seen only in particles smaller than 2 nm where localization of energy levels can be observed when the spacing between the levels exceeds thermal energy (about 26 meV).

Quantum Size Effect

When the size of a nanocrystal becomes smaller than the de Broglie wavelength, electrons and holes get spatially confined, electrical dipoles get generated, the discrete energy levels are formed.

As the size of the material decreases, the energy separation between adjacent levels increases. The density of states of nanocrystals is positioned in between discrete (as that of atoms and molecules) and continuous (as in crystals).

Quantum size effect is most significant for semiconductor nanoparticles. In semiconductors, the bandgap energy is of the order of a few electron volts. It increases with a decrease in particle size.

When photons of light fall on a semiconductor, the photons are absorbed. A sudden rise in absorption is observed when the photon energy is equal to the bandgap.

As the size of the particle decreases, absorption shifts towards the shorter wavelength (blue shifts). This indicates an increase in the bandgap energy (Fig.5.8). A change in absorption causes a change in the colour of the semiconductor nanoparticle.

For example, bulk cadmium sulfide is orange in colour and has a bandgap of 2.42 eV. It becomes yellow and then ultimately white as its particle size decreases and the bandgap increases.

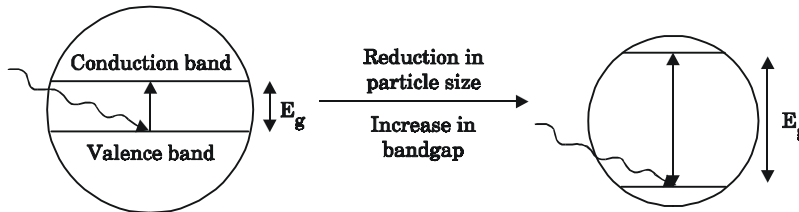


Fig. 5.8 *Increases in bandgap with a decrease in particle size (blue shift)*

5.4 TUNNELING - SINGLE ELECTRON PHENOMENA

The phenomenon of penetration of charge carriers directly through the potential barrier, instead of climbing over it, is called tunneling.

In electronics, transistor is the most important device. Transistors are what computers use to compute-tiny switches turning ON and OFF making logic decisions.

Today, microchips have over a billion transistors, each one turning ON and OFF a billion times every second.

These chips require manufacturing processes with roughly sub 100-nanometer resolution. Every year, this technology resolution drops enabling even smaller transistors. Thus, more transistors are squeezed into the same amount of semiconductor space.

Interestingly, when each transistor is reducing to a few atoms, or a single molecule, quantum effects will play a significant role.

In 1970, to switch ON a silicon transistor required about 10 million electrons. Present day, transistors require closer to 10,000 electrons. Rather than moving many electrons through transistors, it may very well be practical and necessary to move electrons one at a time.

The single electron devices are sensitive to the transfer of even single electron change. Single electron devices provide a potential application of ultra large scale integrated circuits with device size in the order of nanometres. They exhibit high speed operation with lower power dissipation.

5.5 COULOMB – BLOCKADE EFFECTS

As the size of the quantum dot decreases, the charging energy W_c of a single excess charge on the dot increases.

If the quantum-dot size is sufficiently small and the charging energy W_c is much greater than thermal energy $k T$, there is no electron tunnels to and from the quantum dot.

Thus, the electron number in the dot takes a fixed value, say zero, when both the electrodes are grounded.

Definition

The charging effect which blocks the injection or rejection of a single charge into or from a quantum dot is called *Coulomb blockade effect*.

Condition for coulomb blockade

If two or more charges near one another, they exert coulomb forces upon each other. If two charges are the same kind, the force is repulsive. Therefore, the condition for observing coulomb blockade effect is expressed as

$$W_c = \frac{e^2}{2C} \gg k T$$

where C – capacitance of the quantum dot

T – temperature of the system.

W_c – charging energy and this is the energy needed to add one negatively charged electron to the dot.

Single Electron Tunneling

Tunneling is the way the electrons cross both the physical barriers and the energy barriers separating a quantum dot from the bulk material that surrounds it.

If any number of electrons on one side of the barrier could just tunnel across it, there would not be any isolation.

So it is necessary to control the addition and removal of electrons in a quantum dot.

When the size gets reduced, the capacitance also reduces to a small value.

At small sizes, the energy required to store an additional electron on it, $W = \frac{Q^2}{2C}$, may become larger than the thermal energy kT .

As a consequence, the quantization of charge can dominate and tunneling of single electrons across leaky capacitors carries the current. **This is called single electron tunneling (Fig. 5.9).**

It is used to design new types of devices, for example single-electron transistor using quantum dots.

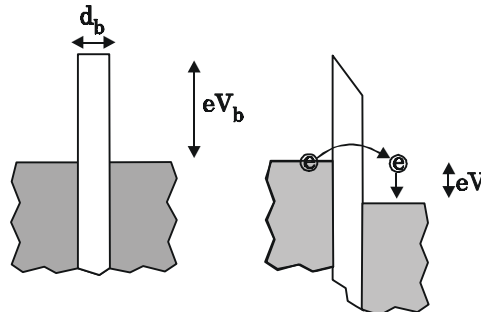


Fig. 5.9 Schematic representation of an electron tunneling through a barrier

So single-electron devices are devices that can control the motion of even a single electron and consist of quantum dots which have tunnel junctions.

5.6 SINGLE ELECTRON TRANSISTOR (SET)

Although a single-electron box can control the number of electrons in the quantum dot, it does not have the properties of a switching device.

Single switching devices are essential elements in **Ultra large scale integrated circuits (ULSIs)**. The single-electron switching devices are required to utilize single electron devices in logic circuits.

Definition

SET is three-terminal switching device which can transfer electrons from source to drain one by one.

That is the purpose of the SET is to individually control the tunneling of electrons into and out of the quantum dot.

Construction & working

The structure of SET is shown in fig.5.10. It has the similar structure of a conventional Field Effect Transistor (FET).

It has tunneling junctions in place of pn-junctions and quantum dot in place of the channel region of the FET.

To control tunneling, a voltage bias to the gate electrode is applied. A separate voltage bias is applied between source and drain electrodes for the current direction.

For current to flow, gate bias voltage must be large enough to overcome the Coulomb blockade energy.

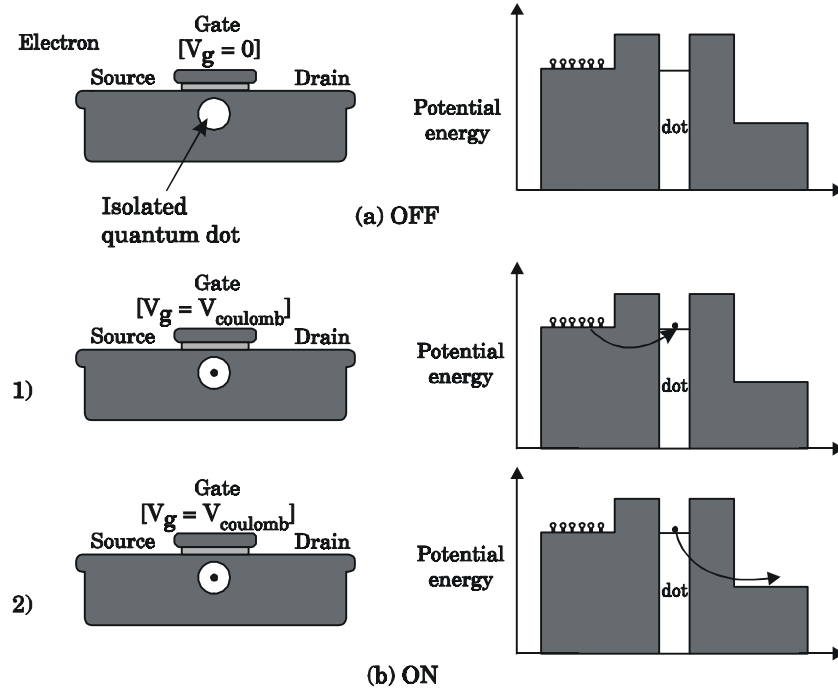


Fig. 5.10 A single-electron transistor (SET)

(a) SET in "OFF" mode.

(b) SET in "ON" mode.

The energy E needed to move a charge Q , across a potential difference V is given by

$$E = VQ.$$

Presently, the charge of an electron ($Q = e$). So, the voltage that will move an electron onto or off the quantum dot is given by

$$\therefore V = \frac{E}{e} = \frac{W_c}{e}$$

Since $E = W_c$,

where, W_c – charging energy

$$V = \frac{\frac{e^2}{2C}}{e} = \frac{e}{2C} \quad \dots(1) \quad \left[\because W_c = \frac{e^2}{2C} \right]$$

With this voltage applied to the quantum dot, an electron can tunnel through Coulomb blockade of the quantum dot. The gate voltage as given in eqn.(1) can be used to tune the number of electrons on the dot at a time.

At the lowest setting, electrons tunnel one at a time, from source to drain through the dot. At the proper gate voltage, $V = \frac{e}{2C}$ the potential energy of the dot is low enough to allow an electron to tunnel through Coulomb blockade energy barrier to the quantum dot.

Once the quantum dot gets an electron, its potential energy rises. Then, the electron tunnels through the Coulomb blockade on the other side reach the lower potential energy at the drain. Since, the dot is empty and potential is lower again, the process repeats.

Thus, when the gate voltage V_g is zero, no current flows. The first gate voltage which is large enough to move an electron through the Coulomb blockade is called V_{Coulomb} .

For single-electron tunneling, $V_g = V_{\text{Coulomb}}$.

If the gate voltage equals $V_{\text{Coulomb}} + \frac{e}{2C}$, then two electrons can be moved on the quantum dot at a time. If the gate voltage equals

$$V_{\text{Coulomb}} + \frac{e}{2C} + \frac{e}{2C} = V_{\text{Coulomb}} + \frac{e}{C},$$

then three electrons can be moved on the quantum dot at a time and so on.

Thus, number of electrons in the quantum dot is controlled using the gate voltage.

These ON and OFF states can be utilized to make an effective switch out of a SET.

The gate voltages for a SET are a few millivolts and source-to-drain currents are in the pico ampere range.

Advantages

- The fast information transfer speed between cells (almost near light speed) is carried out via electrostatic interactions only.
- No wire is needed between arrays. The size of each cell can be as small as 2.5 nm. This made them very suitable for high density memory.
- This can be used for the next generation quantum computer.

Limitations

- In order to operate SET circuit at room temperature, the size of the quantum dot should be smaller than 10 nm.
- It is very hard to fabricate by traditional optical lithography and semiconductor process.
- The methods must be developed for connecting the individual structures into logic circuits and these circuits must be arranged into larger 2D patterns.

Applications

- The main fields of application of the single electron transistor is used in sensor technology and digital electronic circuits.
- A variety of digital logic functions, including AND or NOR gates, is obtained based on SET operating at room temperature.
- It is used for mass data storage.
- It is used in highly sensitive electrometer.
- SET can be used as a temperature probe, particularly in the range of very low temperatures.
- SET is a suitable measurement set-up for single electron spectroscopy.

- It is used for the fabrication of a homo-dyn receiver operating at frequencies between 10 and 300 MHz.

Concept of Resonant Tunneling

An interesting phenomena occurs when two barriers of width a separated by a potential well of small distance L as shown in fig. 5.11. This leads to the concept of *resonant tunneling*.

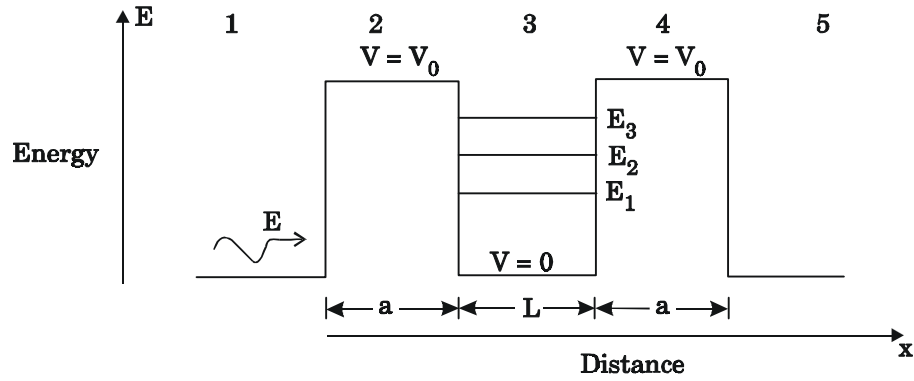


Fig. 5.11 Double barrier junction with no applied bias. E is the energy of the incident electron and E_1, E_2 are the energy level in quasi band states in the well.

For example, assume that incident electrons have energy E and that all the energy states E_n lie above E , as shown in fig. 5.11.

The barriers are sufficiently thin to allow tunneling and the well region between the two barriers is also sufficiently narrow to form discrete (quasi-bound) energy levels, as shown in fig. 5.11.

The transmission coefficient of the double symmetric barrier becomes unity (ie., $T = 1$), when the energy of the incoming electron wave (E) coincides with the energy of one of the discrete states formed by the well.

$$\text{ie., } E = E_n = \frac{n^2 h^2}{8 m L^2}$$

where $n = 1, 2, 3 \dots$

Thus, transmission probability of the double symmetric barrier is maximum and hence, the tunneling current reaches peak value when the energy of electron wave is equal to quantised energy state of the well.

This phenomenon is known as resonance tunneling.

- The double barrier junction has important applications to a device known as Resonant Tunneling Diode (RTD)

5.7 RESONANT TUNNELING DIODE

- It is a device that has two tunneling junctions. Its I-V characteristic shows negative differential resistance characteristic.

Definition

A resonant tunneling diode (RTD) is a diode with resonant tunneling structure. The electrons can tunnel through some resonant states at certain energy levels.

Principle

When electron (wave) incident with energy equal to energy level of a potential well of thin barrier, then the tunneling reaches its maximum value. This is known as resonant tunneling.

Structure of RTD

A typical resonant tunneling diode structure is made by using *n*-type GaAs for the regions to the left and right of both barriers (regions 1 and 5) (Fig. 5.12).

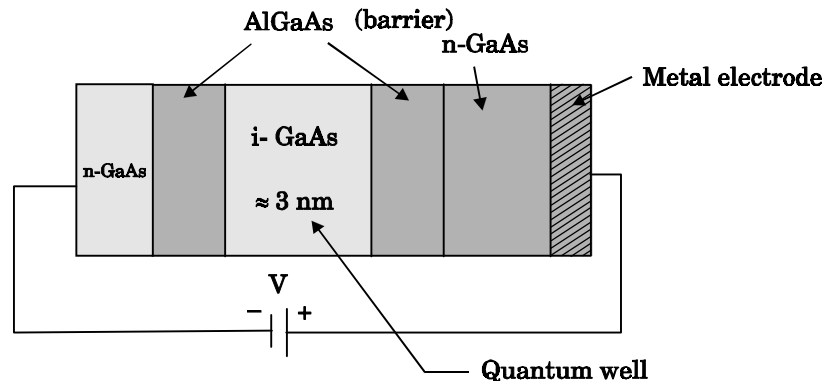


Fig. 5.12 Structure of Resonant Tunneling Diode (RTD)

UNIT V

NANOELECTRONIC DEVICES

Introduction - electron density in bulk material – Size dependence of Fermi energy– quantum confinement – quantum structures - Density of states in quantum well, quantum wire and quantum dot structures – Zener-Bloch oscillations – Resonant tunneling – Quantum interference effects – Mesoscopic structures: Conductance fluctuations and coherent transport – Coulomb blockade effects - Single electron phenomena and Single electron Transistor – Magnetic semiconductors– spintronics - Carbon nanotubes: Properties and applications.

INTRODUCTION

- Nanoscience is a branch of science which deals with the study of phenomena and manipulation of materials at nanometer scales.
- Nanotechnology is the design, production, characterization and application of structures, devices and systems by controlling shape and size at the nanometer scale.
- **Definition:** Nanomaterials are newly developed materials with grain (particle) size at the nanometer range ($1\text{nm} = 10^{-9}$) ie., in the order of 1 – 100 nm.
- Nano devices are the devices which consists of nano materials, its efficiency is more.
- Nano electronics refers to the use of nanotechnology in electronic components, especially transistors.
- The unique micro structure provides tremendous strength, hardness, formability and toughness. to this is has tremendous strength, hardness, toughness and high magnetic & chemical properties.
- (Ex: nano copper is 5 times more stronger than ordinary copper)

ELECTRON DENSITY IN BULK MATERIAL

- **Definition:** Electron density is the number of electrons per unit volume in a material. It is determined by using density of energy states.
- In a solid, the total number of electron energy states ' N ' with energies upto ' E ' is determined based on quantum mechanics using the following equation.

N

$$= \left(\frac{8\pi}{3}\right) (2mE)^{3/2} \left(\frac{a^3}{h^3}\right) \quad \text{--- --}$$

-- (1)

- Here, ' a^3 ' → Volume of the material (' a ' is the characteristic dimension of the solid)

E → Maximum energy level m → Mass of an electron h → Planck's constant.

∴ Number of energy states per unit volume, $n = \frac{N}{a^3}$

$$= \left(\frac{8\pi}{3}\right) \frac{(2mE)^{3/2}}{h^3} \quad \text{--- -- (2)}$$

- Density of states function, $D(E)$ is obtained after taking the derivative of this expression with respect to energy $\left(\frac{dn}{dE}\right)$.

Number of energy states per unit volume per unit energy, $D(E) = \frac{dn}{dE}$

$$\begin{aligned} &= \frac{d}{dE} \left[\frac{8\pi}{3} \frac{(2mE)^{3/2}}{h^3} \right] \\ &= \frac{8\pi}{3} \frac{(2m)^{3/2}}{h^3} \frac{d(E)^{3/2}}{dE} \\ &= \frac{8\pi}{3} \frac{2^{3/2} m^{3/2}}{h^3} \left(\frac{3}{2} E^{\frac{3}{2}-1} \right) \\ &= \frac{8\pi}{3} \frac{2^1 \times 2^{1/2} m^{3/2}}{h^3} \times \left(\frac{3}{2} E^{1/2} \right) \end{aligned}$$

$$D(E) = \frac{8\pi \sqrt{2} m^{3/2}}{h^3} \times \sqrt{E} \quad \text{--- (3)}$$

- In a conductor at '0 K', the electron distribution goes from zero energy upto Fermi energy ' E_F '
- So, the number of free electrons per unit volume or electron density in a bulk conductor at '0 K'

$$n_e = \frac{8\pi}{3} \left(\frac{(2mE_F)^{3/2}}{h^3} \right) \quad \text{--- (4)}$$

SIZE DEPENDENCE OF FERMI ENERGY

Fermi energy:

It is defined as the highest energy level occupied by the electron at '0K' in metal

$$\text{Fermi energy, } E_F = \frac{h^2}{2m} \left(\frac{3n_e}{8\pi} \right)^{2/3}$$

- Here, n_e is the only variable and all the other terms are constants.
- So, we can write, $E_F \propto n_e$
- Thus, Fermi energy of the conductor just depends on the number of free electrons per unit volume (electron density).
- As the electron density is a property of the material, Fermi energy does not vary with the material's size.
- E_F is same for small volume and large volume materials. Also it is same for small size and large size materials.
- But, the average spacing between energy states is inversely proportional to the volume of the solid.

$$\Delta E \propto \frac{1}{V} \quad (\text{or}) \quad \Delta E \propto \frac{1}{a^3}$$

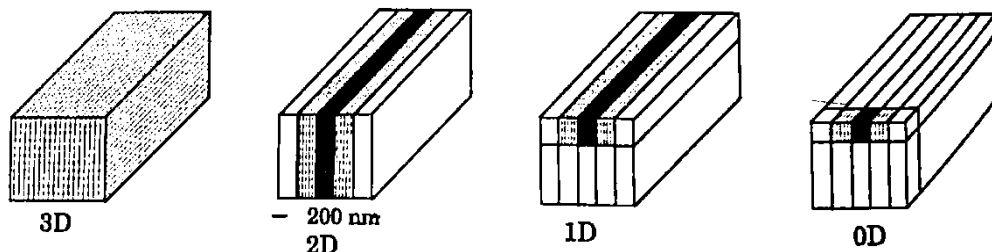
QUANTUM CONFINEMENT

Definition: It is a process of reduction of the size of the solid such that the energy levels inside becomes discrete.

- In this case, small “droplets” of isolated electrons are created.
- Hence, it is possible for us to count few electrons.
- This effect is observed when the size of the particle is too small.
- If we decrease the size of the particle to nano size, the decrease in confining size creates the energy levels discrete.
- The formation of discrete energy levels increases or widens up the band gap and finally the band gap energy also increases
- Since, the band gap and wavelength are inversely related to each other the wavelength decreases with decrease in size and the proof is the emission of blue radiation.

QUANTUM STRUCTURES

- A quantum confined structure is one in which the motion of the electrons or holes are confined in one or more directions by potential barriers.
- Based on the confinement direction, quantum confined structure will be classified into three categories as quantum well, quantum wire and quantum dot.



Types of structure

1. Quantum well(2D): When the electrons are confined in one dimension

then the structure is known as quantum well

2. Quantum wire(1D): When two dimensions are minimized, then the structure is known as quantum wire.

3. Quantum dots(0D): When three dimensions are minimized then it is quantum wire

The classification is shown below in table.

Structure	Quantum Confinement	Number of dimensions
Bulk	0	3 (x, y, z)
Quantum well	1 (z)	2 (x, y)
Quantum wire	2 (x, y)	1 (z)
Quantum dot	3 (x, y, z)	0

DENSITY OF STATES IN QUANTUM WELL

The two dimensional density of states is the number of states per unit area and unit energy.

Let us consider an electron in a two dimensional potential well. The electron is confined in the well.

We have to find out number of energy levels available in between E and E+dE

The space consists of only X-Y plane

In 2 dimensional space $n^2 = n_x^2 + n_y^2$

No. of energy states within the circle of radius 'n'

$$= \pi n^2 \quad \text{--- (1)}$$

No. of energy states within the circle of radius ' $n + dn$ '

$$= \pi(n + dn)^2 \quad \text{--- (2)}$$

The quantum numbers (n_x and n_y) can have only positive integer values. It will have positive values

at $\frac{1}{4}$ part of the circle.

No. of energy states within the circle of radius ' n ' = $\frac{1}{4} \times \pi n^2$

$$= \frac{1}{4} \pi n^2 \quad \text{--- (3)}$$

No. of energy states within the circle of radius ' $n + dn$ ' = $\frac{1}{4} \times \pi(n + dn)^2$

$$= \frac{1}{4} \pi(n + dn)^2 \quad \text{--- (4)}$$

No. of energy states between energy interval E and $E + dE$

$$Z(E) dE = \frac{1}{4} \pi (n + dn)^2 - \frac{1}{4} \pi n^2$$

$$Z(E) dE = \frac{1}{4} \pi [(n + dn)^2 - n^2]$$

$$Z(E) dE = \frac{1}{4} \pi (n^2 + 2ndn + dn^2 - n^2) \quad \text{--- (5)}$$

Since ' dn ' is very small the higher orders of ' dn ' can be neglected.

$$Z(E) dE = \frac{\pi}{4} \times 2ndn$$

$$Z(E) dE = \frac{\pi}{2} n dn \quad \text{--- (6)}$$

The energy of an electron in a cubical metal piece of sides ' a ' is given by, E

$$= \frac{n^2 h^2}{8ma^2} \quad \text{--- (7)}$$

$$n^2 h^2 = 8ma^2 E$$

$$n^2 = \frac{8ma^2 E}{h^2} \quad \text{--- (8)}$$

$$n = \left[\frac{8mE}{h^2} \right]^{\frac{1}{2}} a \quad \text{--- (9)}$$

differentiating equation (9) with respect to 'n' we get,

$$dn = \left[\frac{8m}{h^2} \right]^{\frac{1}{2}} a \frac{1}{2} E^{\frac{1}{2}-1} dE \quad \text{--- -- (10)}$$

$$dn = \frac{1}{2} \left[\frac{8m}{h^2} \right]^{\frac{1}{2}} a E^{-\frac{1}{2}} dE \quad \text{--- -- (11)}$$

Substituting (9) & (11) in (6) we get,

$$Z(E)dE = \frac{\pi}{2} \left[\frac{8mE}{h^2} \right]^{\frac{1}{2}} a \frac{1}{2} \left[\frac{8m}{h^2} \right]^{\frac{1}{2}} a E^{-\frac{1}{2}} dE \quad \text{--- -- (12)}$$

$$Z(E)dE = \frac{\pi}{4} \left(\frac{8m}{h^2} \right) a^2 E^0 dE \quad \text{--- -- (13)}$$

Pauli's exclusion principle states that "Two electrons of opposite spin can occupy each state". Hence, the number of energy states available for electron occupancy is given by,

$$Z(E)dE = 2 \times \frac{\pi}{4} \left(\frac{8m}{h^2} \right) a^2 E^0 dE \quad \text{--- -- (14)}$$

$$Z(E)dE = \frac{\pi}{2} \left(\frac{8m}{h^2} \right) a^2 E^0 dE \quad \text{--- -- (15)}$$

The number of energy states per unit area per unit energy is

$$Z(E) = \frac{\pi}{2} \left(\frac{8m}{h^2} \right) E^0 \quad (\because a^2 = 1 \text{ for unit area})$$

$$Z(E) \propto E^0$$

$$\text{i.e. } Z(E)^{2D} \propto E^0 = \text{constant}$$

This is the expression for density of charge carriers in the energy interval E and E+dE in a quantum well

DENSITY OF STATES IN QUANTUM WIRE

The one dimensional density of states is the number of states per unit length and unit energy.

Let us consider an electron in a one dimensional quantum wire. We have to find out number of energy

levels available in between E and E+dE

The space consists of only X plane

In 1 dimensional space $n^2 = n_x^2$

No. of energy states in the wire of length 'n' = n — — — — — (1)

No. of energy states in the wire of length 'n + dn'

$$= n + dn \quad \text{--- --- (2)}$$

The quantum numbers (n_x) can have only positive integer values. It will have positive values at $\frac{1}{2}$ part of the wire.

$$\begin{aligned} \text{No. of energy states within the wire of length 'n'} &= \frac{1}{2} \times n \\ &= \frac{1}{2} n \quad \text{--- --- (3)} \end{aligned}$$

$$\begin{aligned} \text{No. of energy states within the wire of length 'n + dn'} &= \frac{1}{2} \times (n + dn) \\ &= \frac{1}{2} (n + dn) \quad \text{--- --- (4)} \end{aligned}$$

No. of energy states between energy interval E and E + dE

$$\begin{aligned} Z(E) dE &= \frac{1}{2} (n + dn) - \frac{1}{2} n \\ Z(E) dE &= \frac{1}{2} [n + dn - n] \\ Z(E) dE &= \frac{1}{2} (dn) \quad \text{--- --- (5)} \end{aligned}$$

The energy of an electron in a cubical metal piece of sides 'a' is given by, E

$$= \frac{n^2 h^2}{8ma^2} \quad \text{--- -- -- -- (6)}$$

$$n^2 h^2 = 8ma^2 E$$

$$n^2 = \frac{8ma^2 E}{h^2} \quad \text{--- -- -- -- (7)}$$

$$n = \left[\frac{8mE}{h^2} \right]^{\frac{1}{2}} a \quad \text{--- -- -- -- (8)}$$

Differentiating equation (11) with respect to 'n' we get,

$$dn = \left[\frac{8m}{h^2} \right]^{\frac{1}{2}} a \frac{1}{2} E^{\frac{1}{2}-1} dE \quad \text{--- -- -- -- (9)}$$

$$dn = \frac{1}{2} \left[\frac{8m}{h^2} \right]^{\frac{1}{2}} a E^{-\frac{1}{2}} dE \quad \text{--- -- -- -- (10)}$$

Substituting (10) in (5) we get,

$$Z(E)dE = \frac{1}{2} \frac{1}{2} \left[\frac{8m}{h^2} \right]^{\frac{1}{2}} a E^{-\frac{1}{2}} dE \quad \text{--- -- -- -- (11)}$$

$$Z(E)dE = \frac{1}{4} \left[\frac{8m}{h^2} \right]^{\frac{1}{2}} a E^{-\frac{1}{2}} dE \quad \text{--- -- -- -- (12)}$$

Pauli's exclusion principle states that "Two electrons of opposite spin can occupy each state".

Hence, the number of energy states available for electron occupancy is given by,

$$Z(E)dE = 2 \times \frac{1}{4} \left[\frac{8m}{h^2} \right]^{\frac{1}{2}} a E^{-\frac{1}{2}} dE \quad \text{--- -- -- -- (13)}$$

$$Z(E)dE = \frac{1}{2} \left[\frac{8m}{h^2} \right]^{\frac{1}{2}} a E^{-\frac{1}{2}} dE \quad \text{--- -- -- -- (14)}$$

The number of energy states per unit length per unit energy is

$$Z(E) = \frac{1}{2} \left[\frac{8m}{h^2} \right]^{\frac{1}{2}} E^{-\frac{1}{2}} \quad (\because a = 1 \text{ for unit length})$$

$$Z(E) \propto E^{-\frac{1}{2}}$$

$$\text{i.e. } Z(E)^{1D} \propto E^{-\frac{1}{2}}$$

This is the expression for density of charge carriers in the energy interval E and $E+dE$ in a quantum wire.

DENSITY OF STATES IN QUANTUM DOT

In quantum dot, the electron is confined in all three dimensions and hence no motion of electron is possible. So the density of states is merely a delta function.

$$Z(E)^{1D} = 2\delta$$

2 accounts for spin

Type of structures & No. of non confinement diemensions	Density of state function
Bulk (3D)	$D(E) = \frac{8\pi \sqrt{2} m^{*3/2} (E - E_c)^{1/2}}{h^3}$
Quantum well (2D)	$D(E) = \frac{4\pi m^*}{h^2} \quad E > E_i, \quad i = 1, 2, 3$
Quantum wire (1D)	$D(E) = \frac{2 \sqrt{2} m^* (E - E_i)^{-1/2}}{h} \quad ; \quad i = 1, 2, 3$
Quantum dot (0D)	$D(E) = \delta (E - E_i) \quad ; \quad i = 1, 2, 3$

6.10 DENSITY OF STATES FOR QUANTUM DOT (OR) DENSITY OF STATES IN ZERO DIMENSION

In a zero dimensional system (such as quantum dot), the density of states are truly discrete and they don't form a quasi continuum.

Explanation

In zero dimensional system [quantum dot], the electron is confined in all three spatial dimensions and hence no motion of electron is possible. Each quantum state of a zero dimensional system can therefore be occupied by only two electrons. So, the density of states for a quantum dot is merely a delta function.

$$Z(E)^{0D} = 2\sqrt{(E - E_0)} \quad \dots(1)$$

Here, the factor 2 accounts for spin.

For more than one quantum state, the density of states is given by

$$Z(E)^{0D} = \sum_n 2\delta\sqrt{(E - E_n)} \quad \dots(2)$$

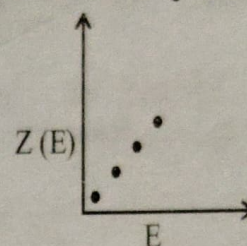
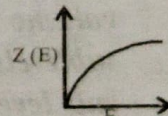
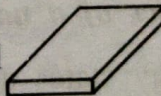
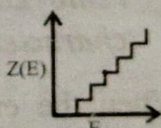
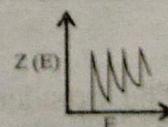


Fig. 6.16 Density of States for a quantum Dot (0-D)

The density of states for quantum dot is shown in Fig 6.16

Table 6.2 Summary of the density of states with various degrees of freedom.

Degrees of freedom	Confinement dimensions	Quantum Structure	Graph for Density of states
3	0	Bulk 	
2	1	Quantum well 	
1	2	Quantum wire 	
0	3	Quantum dot 	